

CSA15 Cloud Computing and Big Data

Analytics Local IaaS Lab: Virtual Disk Allocation and VM Hardware Compatibility

Course Code	CSA15	Course Title	Cloud Computing and Big Data Analytics
Guided Project	GP-VM-02	Submission	Individual evidence through GitHub and Google Classroom
Faculty	Dr C. Rajesh Babu	Employee ID	SSETSCS415
Target Learner	Beginner CSE student	Estimated Duration	2 to 3 lab hours
Primary Tool	Oracle VirtualBox or VMware Workstation	Cloud Link	Connects VM hardware planning with IaaS resource configuration
Guided Project Pattern Read the idea, perform the action, observe the result, record evidence, explain what happened, extend the setup to your capstone context, and submit the final proof through GitHub and Google Classroom.			

Learning Outcomes

- Inspect the existing VM configuration created in the previous guided project.
- Create or plan an additional virtual hard disk using VirtualBox or VMware.
- Record disk controller, disk format, allocated size and storage-allocation method.
- Change or document VM hardware compatibility through clone/create-new-VM planning where supported.
- Explain how virtual CPU, RAM and storage decisions affect capstone infrastructure planning.

Before You Start

- Completed VM from GP-VM-01 using VirtualBox or VMware.
- At least 3 GB free host disk space for an additional test disk, or a written storage-aware plan if the laptop is constrained.
- Guest OS boot proof and hypervisor settings screenshots from the previous guide.
- GitHub account and Google Classroom access.

Quick Learning Notes

- A virtual hard disk is a file on the host system that behaves like a physical disk inside the VM.
- Dynamically allocated disks grow as data is written; fixed-size disks reserve storage earlier and may perform more predictably.
- A disk controller such as SATA, SCSI or NVMe defines how the guest OS communicates with the virtual disk.
- Hardware compatibility controls which virtual hardware version/features the VM can use. In some tools it is changed through clone, export, import or new-VM creation.

Lab Exercise Coverage

Experiment 9	Create a virtual hard disk and allocate storage using the selected hypervisor.
Experiment 12	Change or document hardware compatibility using clone/create-new-VM strategy for an already configured VM.



Hypervisor Choice

Oracle VirtualBox	Choose when the existing VM was created in VirtualBox.	Storage settings, Controller, Add Hard Disk, VDI/size and VM settings evidence.
VMware Workstation	Choose when the existing VM was created in VMware.	VM Settings, Add Hardware, Hard Disk, SCSI/SATA/NVMe choice, VMDK/size evidence.
Safety Rule Do not attach a host physical disk. Add only a new virtual disk file for this lab. Avoid expanding disks beyond available host storage.		

Architecture View

Existing VM	Settings Panel	Virtual Controller	New Virtual Disk	Compatibility Choice	Evidence

This lab focuses on how virtual hardware is modified after a VM already exists. The goal is to understand storage allocation and compatibility decisions without risking the host system.

Guided Build

Stage 1 - Inspect the Existing VM

Learn: Before changing hardware, the current VM state must be recorded.

- Open VirtualBox or VMware Workstation.
- Select the VM created in GP-VM-01.
- Record VM name, selected OS, CPU, RAM, current disk size, disk format and network mode.
- Shut down the VM before changing storage hardware.

Checkpoint The current VM configuration is recorded before adding a disk.

Record 1 - Existing VM baseline evidence

Existing VM: csa15-localvm-192372057-Clone

Guest OS: Tiny Core Linux CorePlus

CPU: 2 virtual CPUs

RAM: 2048 MB

Existing Disk: 20 GB VDI

Disk Format: Normal (VDI)

Disk Allocation: Dynamically allocated differencing storage

Storage Controller: SATA (AHCI)

ISO: CorePlus-current.iso

Network: NAT

VM Status: Powered Off

Observation: The existing VM baseline was recorded before adding a new virtual disk. The VM has a 20 GB VDI system disk attached through the SATA controller and the CorePlus ISO attached through the IDE controller.

Information

Type (Format)Normal (VDI)

Virtual size20.00 GB

Actual size2.00 MB

Storage detailsDynamically allocated differencing storage

LocationC:\Users\HP\VirtualBox VMs\csa15-localvm-192372057-Clone\csa15-localvm-192372057-Clone.vdi

Attached tocsa15-localvm-192372057-Clone (Baseline-Configured-VM)

Encryption key--

Storage

Devices

Controller: IDE

CorePlus-current.iso

Controller: SATA

csa15-localvm-192372057-Clone.vdi

Attributes

NameSATA

TypeAHCI

Port Count1

☐ Use Host I/O

Information

Type (Format)Normal (VDI)

Virtual size20.00 GB

Actual size2.00 MB

Storage detailsDynamically allocated differencing storage

LocationC:\Users\HP\VirtualBox VMs\csa15-localvm-192372057-Clone\csa15-localvm-192372057-Clone.vdi

Attached tocsa15-localvm-192372057-Clone (Baseline-Configured-VM)

Encryption key--

Stage 2 - Open the Storage or Hardware Wizard

Learn: Virtual storage is added from the hypervisor settings, not from the guest OS

installer. - In VirtualBox, open Settings -> Storage, select the controller area and choose Add Hard Disk.

- In VMware, open VM Settings -> Add Hardware -> Hard Disk.

- Choose Create New Virtual Disk.

- Select a safe test size such as 2 GB unless the faculty asks for another value.

- Choose split disk into multiple files if storage portability is preferred, or a single file if the tool/lab requires it.

Checkpoint

The disk creation screen shows a new virtual disk size and format before final confirmation.

Record 2 - Virtual disk wizard evidence

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Selected Hypervisor: Oracle VirtualBox 7.2.14
Installer filename: VirtualBox-7.2.14-174565-Win
Source: Oracle VirtualBox official download page – <https://www.virtualbox.org/wiki/Downloads>
Download date: 07-08-2026
Approximate file size: 169 MB
Observation: The VirtualBox installer was successfully downloaded and is available in the Downloads folder

VirtualBox-7.2.14-174565-Win	07-08-2026 01:36 PM	Application	1,73,887 KB
Oracle_VirtualBox_Extension_Pack-7.2.14	07-08-2026 01:35 PM	VirtualBox Extensi...	19,599 KB

Stage 3 - Allocate and Attach the Virtual Disk

Learn: A newly created virtual disk must be attached to the VM through a controller before the guest OS can use it.

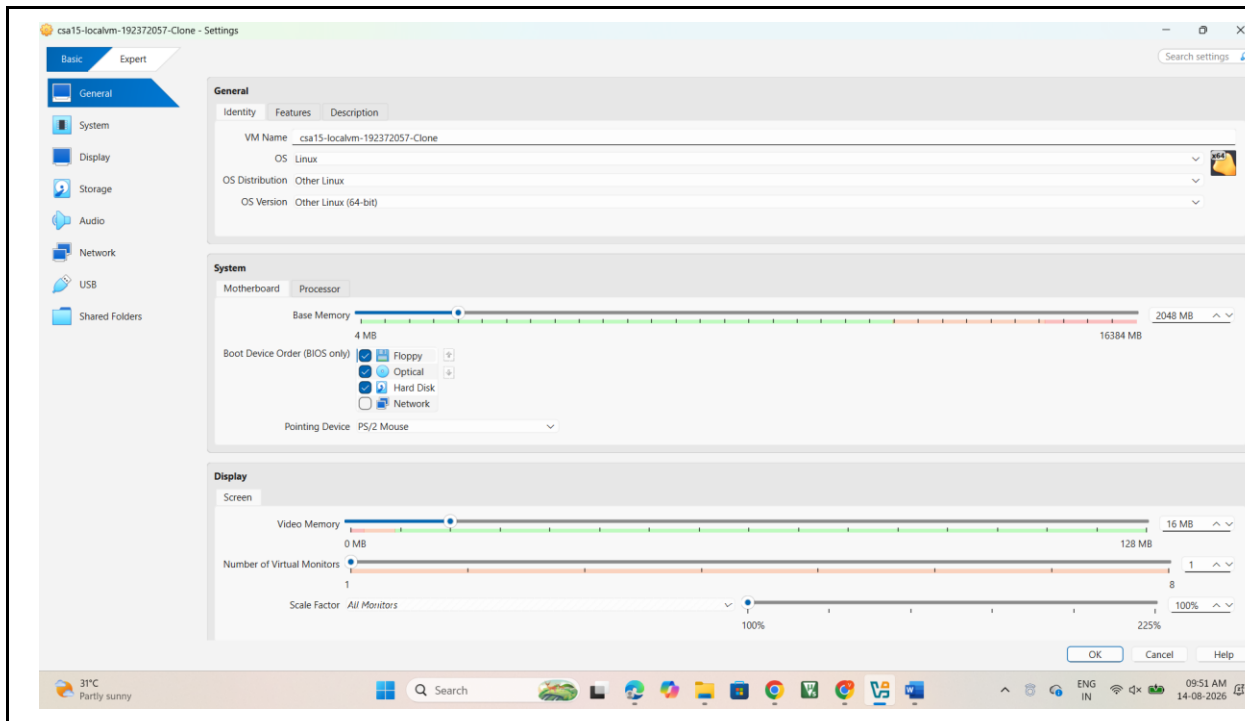
- Confirm the new disk file name using register number, for example csa15-extra-disk-register-number. - Attach the disk to a suitable controller: SATA is acceptable in VirtualBox; SCSI/SATA/NVMe may appear in VMware depending on compatibility.
- Save settings and capture the storage/settings page showing both original disk and new disk.
- Boot the VM and observe whether the guest OS detects an additional disk. Formatting inside the guest OS is optional and should be done only if instructed.

Checkpoint

The hypervisor settings show the additional virtual disk attached to the existing VM.

Record 3 - Attached virtual disk evidence

Hypervisor: Oracle VirtualBox
Version: 7.2.14
Installation Status: Successfully installed
Observation: Oracle VirtualBox was successfully installed on the Windows host system. VirtualBox Manager opens correctly and is used to create and manage the Linux virtual machine.



Stage 4 - Hardware Compatibility Review

Learn: Hardware compatibility controls which virtual hardware features and controller types are available.

- In VMware, inspect VM -> Manage -> Change Hardware Compatibility if available. If it is restricted, record the menu or version limitation.
- In VirtualBox, record equivalent compatibility-related settings such as System, Acceleration, Storage Controller and chipset options.
- If the tool recommends cloning before compatibility change, create a clone plan rather than modifying the only working VM.
- Write which compatibility choice is safest for a beginner lab and why.

Checkpoint

Hardware compatibility evidence or a justified compatibility-change plan is documented.

Record 4 - Hardware compatibility evidence

Selected Guest OS: Tiny Core Linux CorePlus

ISO Filename: CorePlus-current.iso

Approximate ISO Size: About 250 MB

Reason for Selection: Tiny Core Linux CorePlus was selected because it provides a graphical desktop environment while having a relatively small download size. It is suitable for a lightweight VM and reduces bandwidth and resource requirements for the laboratory.

Observation: The CorePlus ISO was attached to the VM and successfully booted to the Tiny Core Linux desktop.

▼ Last week



CorePlus-current

07-08-2026 02:38 PM

Disc Image File

2,82,624 KB

Stage 5 - Resource Impact and Capstone Link

Learn: Extra disk and hardware choices must be justified by workload, not added

randomly. - Record host free disk space before and after creating the additional disk if possible.

- Write the purpose of the added disk: logs, uploads, database files, backup, test data or capstone module storage.
- Compare standard 15 GB VM disk with the extra 2 GB test disk.
- Connect this storage decision to one capstone data requirement.

Checkpoint

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The student explains why the additional disk or compatibility decision is useful.

Record 5 - Resource impact and capstone linkage

VM Name: csa15-localvm-192372057-Clone

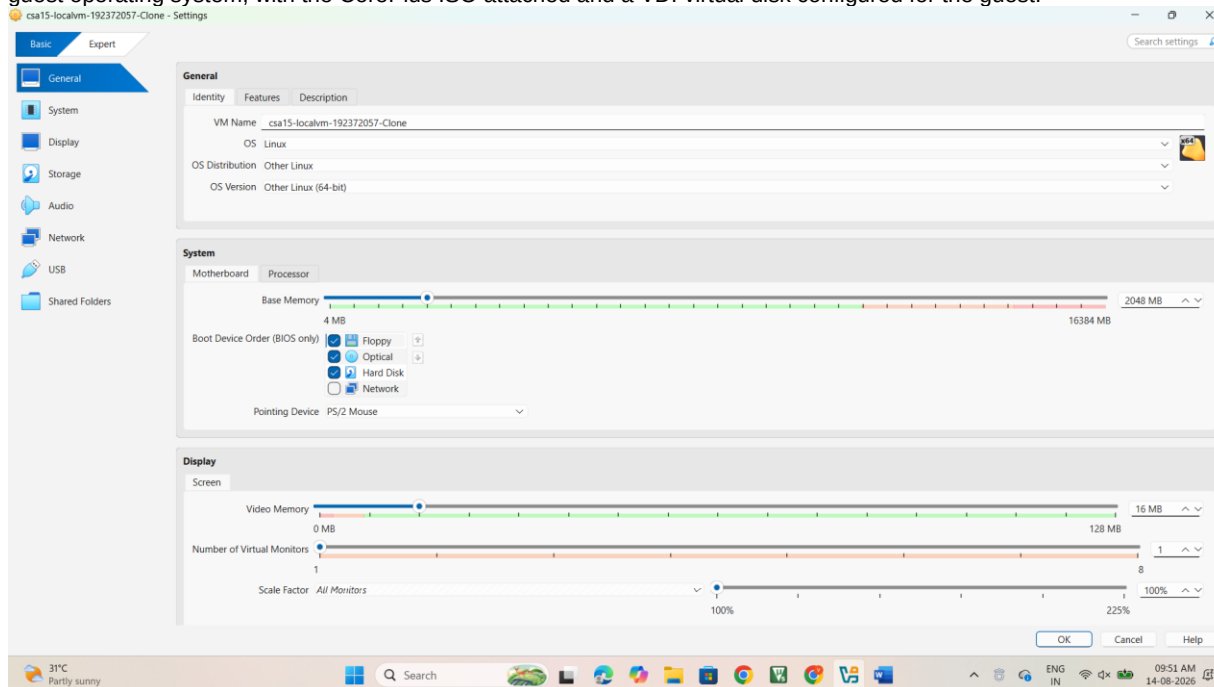
OS Family: Linux

OS Version: Other Linux (64-bit)

ISO Attached: CorePlus-current.iso

Virtual Disk: csa15-localvm-192372057-Clone.vdi

Observation: The Linux virtual machine was successfully created in Oracle VirtualBox. The VM uses Linux as the guest operating system, with the CorePlus ISO attached and a VDI virtual disk configured for the guest.



Final Evidence Checklist

☐	Baseline captured	Existing VM settings before modification.
☐	Additional disk planned/created	Virtual disk wizard screenshot with size and format.
☐	Disk attached	Settings page showing original disk and additional disk.
☐	Compatibility reviewed	VMware compatibility menu or VirtualBox equivalent settings evidence.
☐	Capstone link written	Storage purpose and resource-impact explanation.

Explain and Extend

Explain the difference between a virtual hard disk file and a physical disk.

A virtual hard disk is a file stored on the host computer that behaves like a physical hard disk inside a virtual machine. The guest operating system uses the virtual disk for storing its files and data, while the actual file remains on the host system. A physical disk is real hardware directly connected to the computer.

Explain why dynamically allocated storage is useful for beginner labs.

Dynamically allocated storage is useful because the virtual disk starts with a small amount of actual host storage and grows as data is written to it, up to the configured virtual size. This avoids reserving the entire disk capacity immediately and makes storage management easier for beginner laboratory exercises.

Explain why compatibility changes should be tested on a clone or documented plan.

Compatibility changes should be tested on a clone or documented as a plan because changing virtual hardware compatibility can affect whether the existing VM boots or works correctly. Using a clone protects the original working VM and allows the compatibility change to be tested safely. If cloning is not possible, the intended compatibility change and its expected effect should be documented before modifying the VM.

Extend this setup to your capstone: what data could be stored on a separate disk?

A separate virtual disk could be used to store capstone project data such as application logs, uploaded files, test datasets, database files, backups or generated results. Keeping these files on a separate disk can make data management easier and reduce the risk of affecting the main VM system disk.

Submission Protocol

- Create a GitHub folder named CSA15-GP-VM02-Virtual-Disk-Hardware-Compatibility.
- Upload this completed document as DOCX/PDF along with screenshots and short README.md.
- README.md must include tool used, VM/cloud specifications, configuration decisions, screenshots index and cleanup status.
- Submit the GitHub repository link in Google Classroom. Do not upload passwords, private keys, tokens, public IP credentials or sensitive account screenshots.

Student Assurance

Declaration

I confirm that this guided-project evidence was completed by me, the screenshots are from my own lab environment, and no secret keys, passwords or sensitive account information are included.

Student Signature: K.R.VISHNU CHAITHANYA Register Number: 192372057 Date: 14-08-2026

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